

HOW BIG DATA POWERS A SMART CITY

1

Structured Data

Traffic schedules, Electricity billing records, Water usage logs

2

Semi-Structured Data

Sensor logs (JSON format), GPS tracking data, smart meter readings.

3

Unstructured Data

CCTV footage, emergency voice calls, social media complaints.

4

The 5V's of Big Data

Volume - Massive sensor data

Velocity - Real-time updates

Variety - Text, video, GPS

Veracity - Filtering incorrect data

Value - Smarter city decisions

5

Why Traditional BI Fails

- Supports only structured data
- No real-time processing
- Limited scalability

6

Why Big Data is Essential

- Distributed storage
- Real-time analytics
- Predictive decision-making